

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 17: Circles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Cyclic quadrilaterals

1. A, E, C and B lie in order on a circle. Chords AC and BC are equal, BC is produced to D , and $\angle ACD = 120^\circ$. Find $\angle AEC$. (a) 30° (b) 60° (c) 90° (d) 120° **ICSE 2025, 1 mark**

SOLUTION

$$\angle ACB = 180^\circ - 120^\circ = 60^\circ \text{ (linear pair).}$$

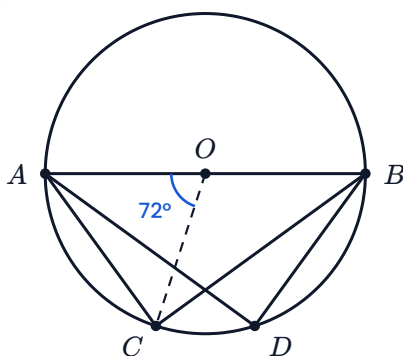
$$AC = BC, \text{ so } \angle ABC = \angle BAC = \frac{180^\circ - 60^\circ}{2} = 60^\circ.$$

$$ABCE \text{ is a cyclic quadrilateral, so } \angle AEC = 180^\circ - \angle ABC = 120^\circ.$$

Answer: (d)

2 Angle at the centre and at the circumference

2. AB is a diameter of a circle with centre O . C and D are points on the circle on the same side of AB , with $AC = BD$ and $\angle AOC = 72^\circ$. Find (a) $\angle ABC$ (b) $\angle BAD$ (c) $\angle ABD$. **ICSE 2020, 3 marks**



SOLUTION

(a) The angle at the circumference is half the angle at the centre on the same arc: $\angle ABC = \frac{72^\circ}{2} = 36^\circ$.

(b) Equal chords subtend equal angles at the centre, so $\angle BOD = 72^\circ$ and $\angle BAD = 36^\circ$.

(c) $\angle ADB = 90^\circ$ (angle in a semicircle), so $\angle ABD = 90^\circ - 36^\circ = 54^\circ$.

Answer: (a) 36° (b) 36° (c) 54°

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu